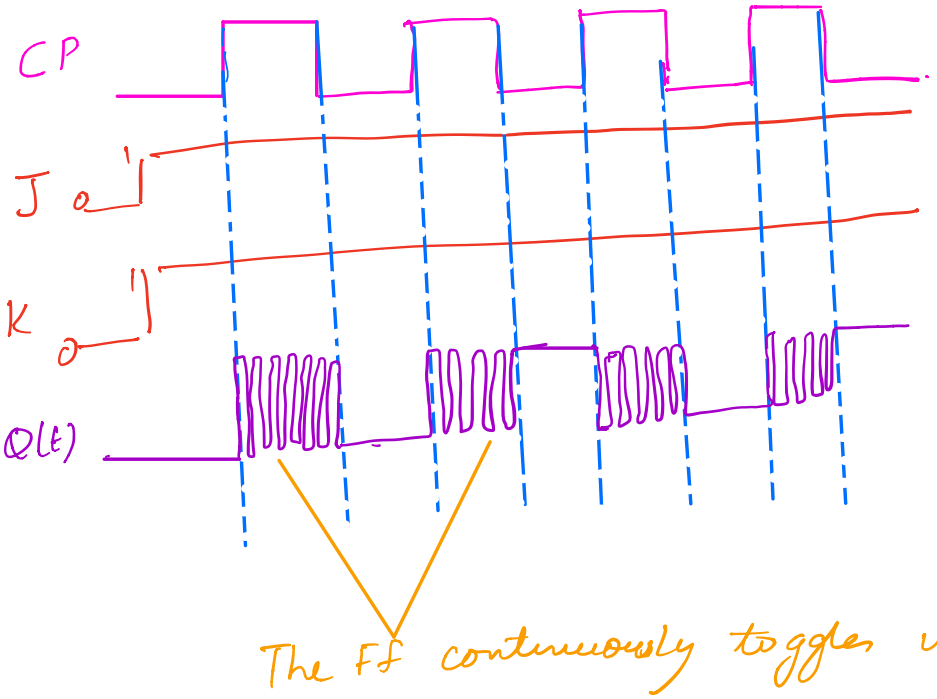
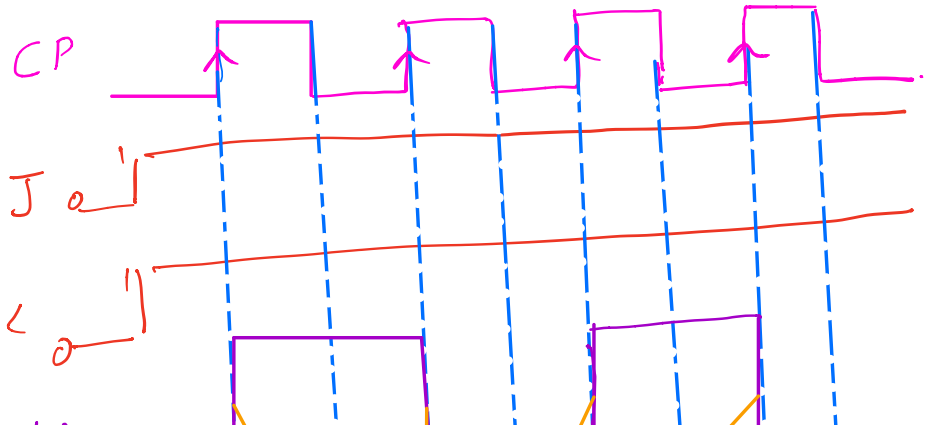
The Problem With Level Triggered FF:

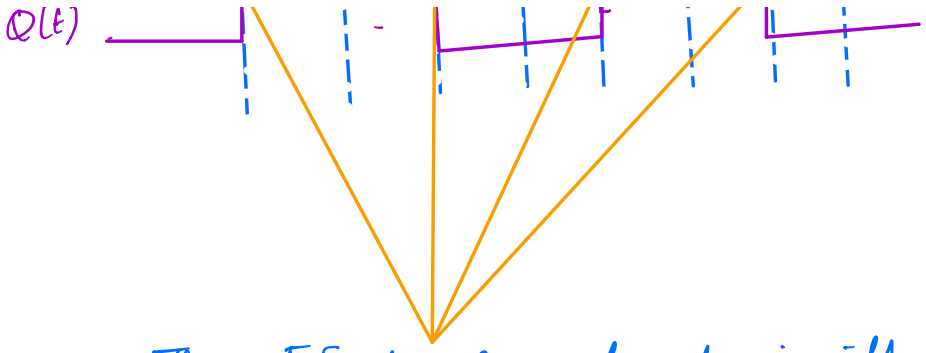
$Q(t+1) = \bar{Q}(t)$, when $J=K=1$ in JK FF, would mean that the o/p changes independent of any change in i/p. In a level triggered FF this results in



The FF continuously toggles when CP=1

We want

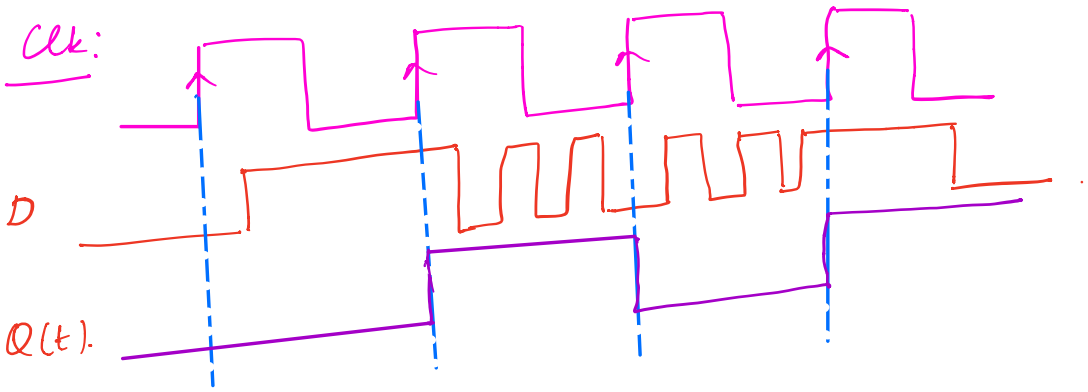




The FF toggles only during the rising edge of the CP or its is Rising Edge (+ve) Triggred FF.

So how does one construct or Design an Edge Triggered Ff ?

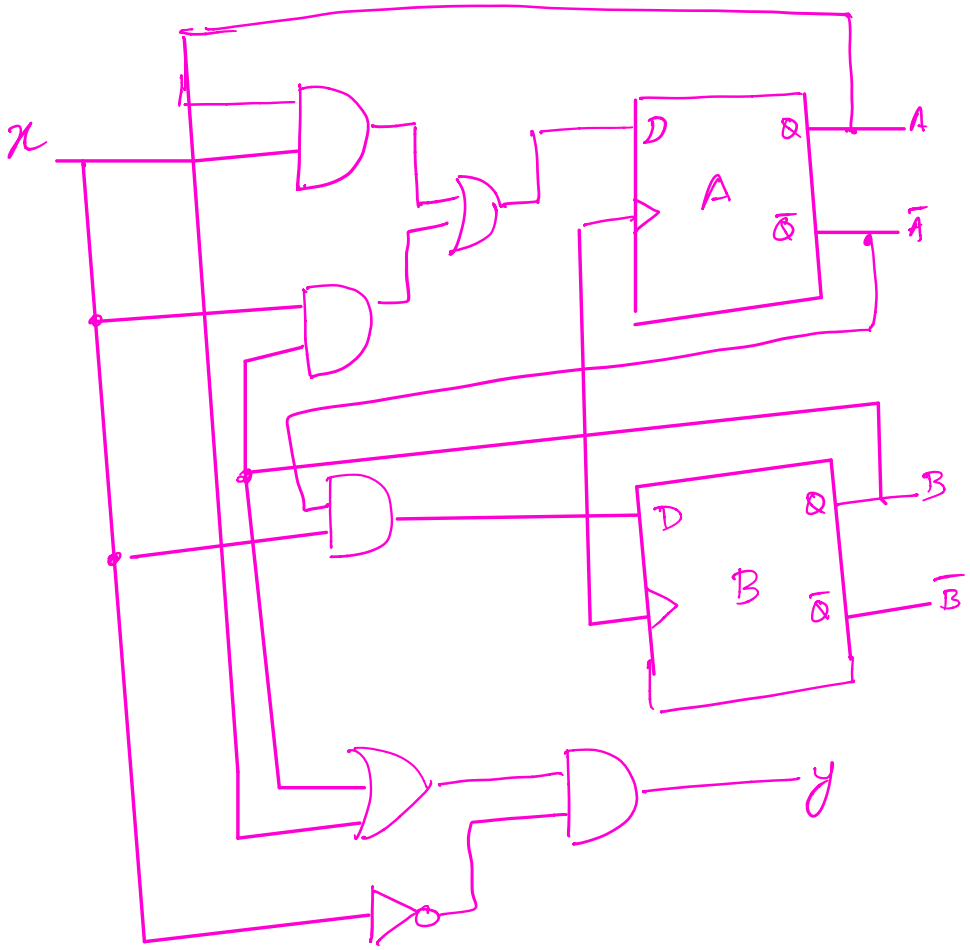
The Timing Diagram for the Edge Triggred D FF:



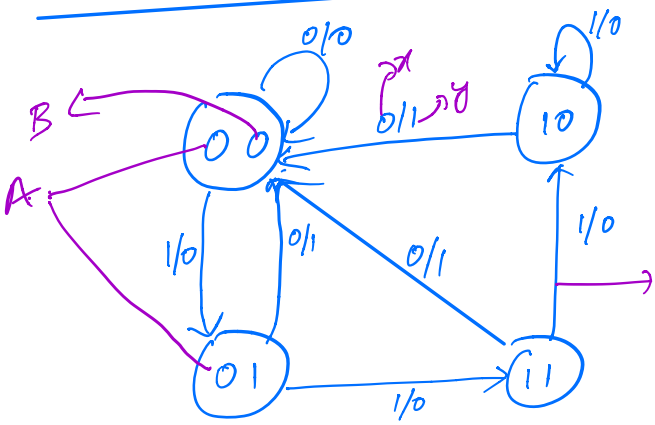
∴ Q changes only at the rising edge. Or Q(t) reflects D's value at the rising edge of the clock, Q (t+1) = D with the rising edge.

Changes to D between two rising edges does NOT affect Q.

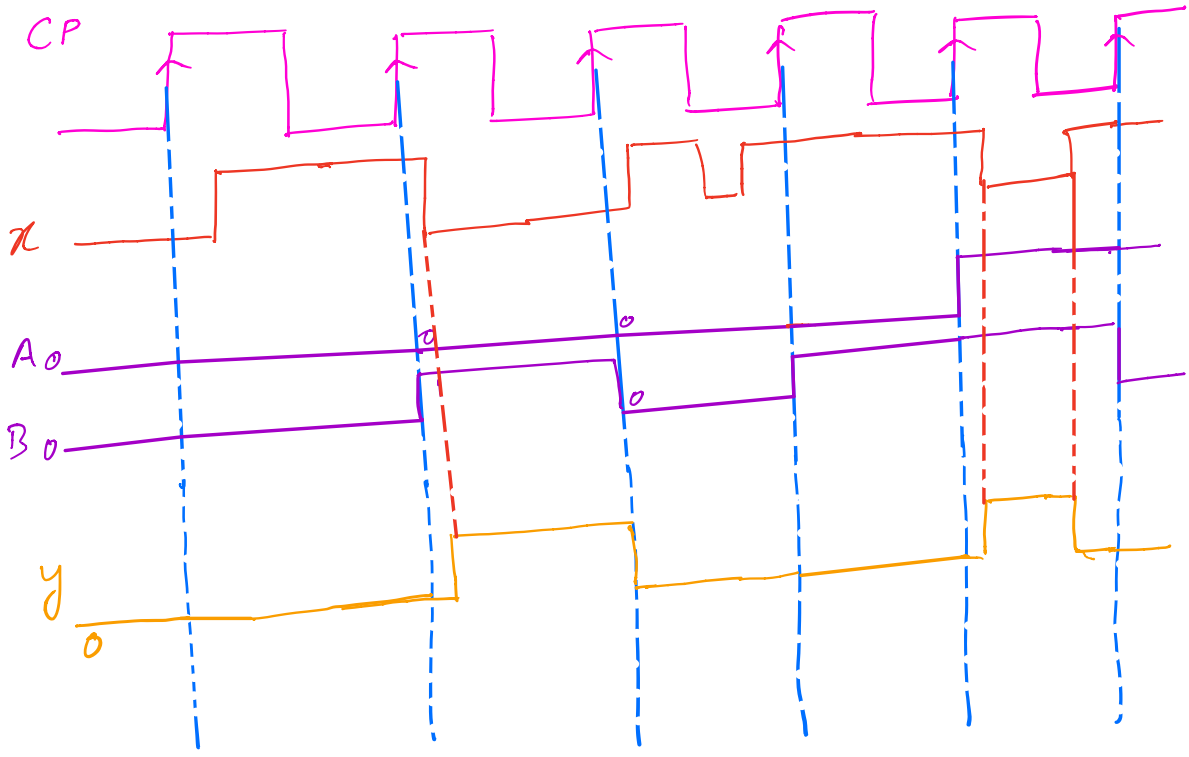
Analysis of Sequential Circuits



FSM for the above ckt



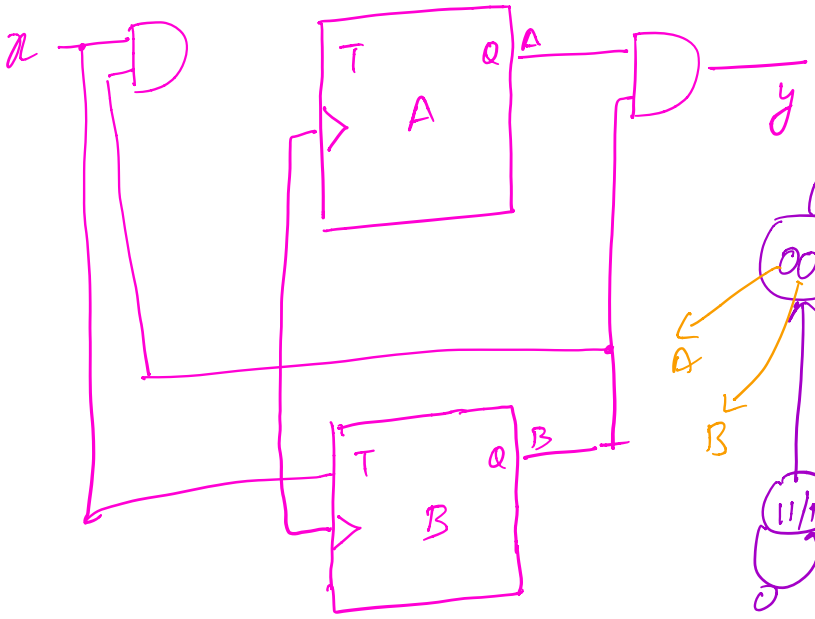
→ Transitions at the rising edge

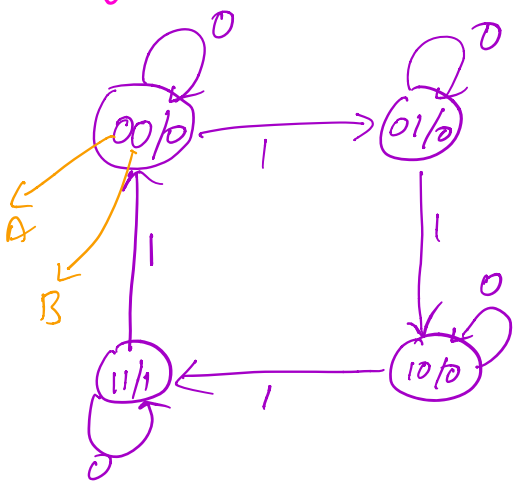


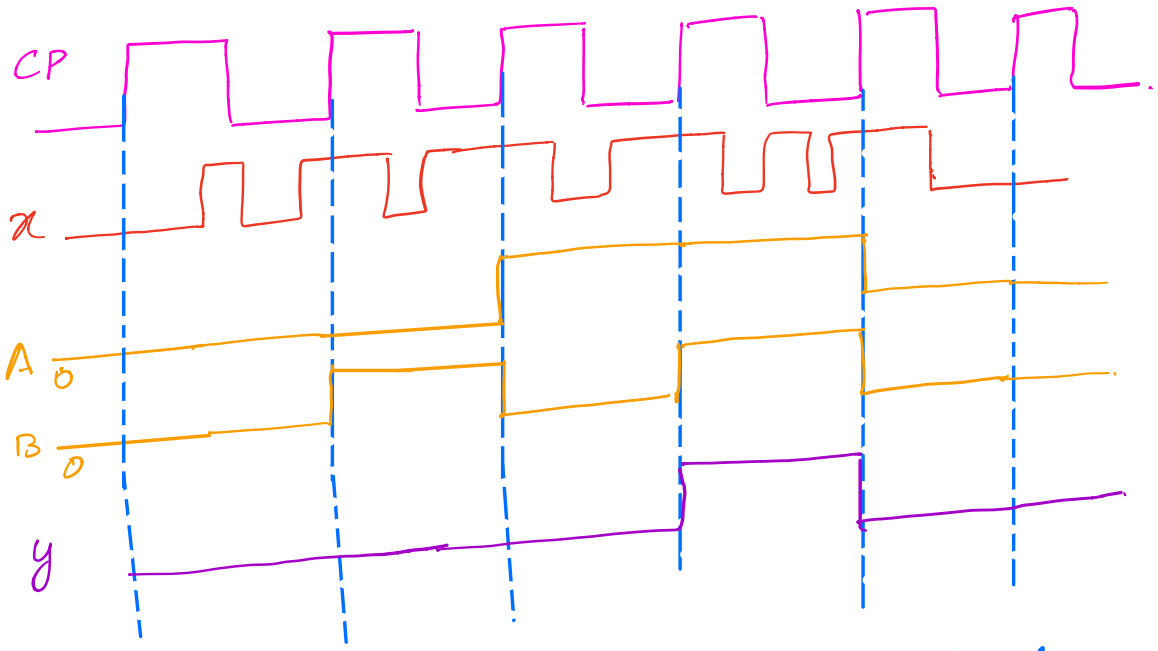
→ Changes in FF are synchronized with the CP.

→ Changes in combinatorial O/P y can occur when the FF changes at the rising edge OR when changes in i/p occur. Changes in y can be Asynchronous with the clock or MEALY FSM or Mealy Seq Ckt.

In contrast







MOORE FSM : Combinatorial o/p y is fn of only the FF states and changes in synch with the CP.